



SE618 LOGISTIC CASE STUDY ANALYSIS

WEEK 11 – PROGRESS REPORT

PROJECT TITLE: SERGE – AI Agent for Supply Chain Operations

Scenario Restructuring, Pipeline Resilience & Evaluation Runs

Gauthier SCHMITT	6822807466
Louis GRIGNOLA	6822807516
Matteo SCHWARZ	6822807557
Maël TELLUS	6822807532

SUBMITTED TO:

AJARN WARUT

APRIL 2026

SEMESTER 2/2025

**SCHOOL OF MANUFACTURING SYSTEMS AND MECHANICAL ENGINEERING
SIRINDHORN INTERNATIONAL INSTITUTE OF TECHNOLOGY
THAMMASAT UNIVERSITY**

Contents

1	Summary	2
2	Use Case Restructuring by Complexity	2
3	Pipeline Resilience	2
4	LLM-as-Judge Tuning	3
5	Evaluation Runs Launched	4
6	Final Report Drafting	4

Summary

This week was focused on **evaluation readiness**: reworking the use cases to reflect realistic operational questions at increasing difficulty, hardening the evaluation pipeline for real-world constraints (quota limits, network issues), tuning the LLM-as-Judge to be fair instead of over-strict, and launching multiple full evaluation passes. We also started writing the final report and prepared the groundwork for aggregating the results.

Use Case Restructuring by Complexity

The previous version of our evaluation had 15 scenarios spread across 3 use cases with a coarse easy/medium/hard tag. After running the agent qualitatively, we realised the agent’s behaviour varies more with **resolution complexity** (how many systems to chain, how much synthesis is needed) than with use case.

We reorganised the evaluation around **5 complexity levels**, each containing real operational questions:

Level	Name	Scenarios	What it tests
L1	Trivial	4	Single lookup in one system
L2	Simple	6	One-system filtering or listing
L3	Moderate	6	Cross-reference between two systems
L4	Complex	6	Causal reasoning across 3–4 systems
L5	Expert	5	Full diagnostic plus actionable recommendation

Total: **27 scenarios**, covering the five business use cases (UC-1 order status, UC-2 root cause, UC-3 supplier sourcing, UC-4 operational monitoring, UC-5 planning) as a second cross-axis.

Each scenario is written as a realistic question an operations manager would ask, and includes a baseline estimate of how long the same task would take a human working manually across multiple systems (ranging from 2 minutes for a trivial lookup to 45 minutes for a full health check).

Pipeline Resilience

Running Claude at scale via the subscription exposed us to two real constraints:

1. **Quota limits** reset periodically and interrupt long evaluation runs.
2. **Transient failures** (network, timeouts) happen occasionally.

We rewrote the evaluation pipeline to be **fully resumable**:

Feature	Behaviour
Idempotent runs	Each run checks if a valid output already exists. Re-running the pipeline skips completed items and continues with pending ones.
Retry with backoff	Failed calls are retried up to 3 times with exponential backoff (15s, 45s, 135s). Handles transient network issues automatically.
Quota detection	Rate-limit and quota errors are recognised by scanning Claude’s output. The pipeline stops cleanly with a clear message instead of burning retries.
Safe interrupt	Ctrl+C exits gracefully, leaving all completed work on disk.
Progress tracking	A <code>-status</code> flag shows how many runs are done without launching anything.
Failure log	Persistent failures are logged with category (<code>quota</code> , <code>network</code> , <code>unknown</code>) for later diagnosis.

Key Insight

The pipeline can now be stopped and resumed across multiple sessions without losing progress. This is essential given the subscription’s 5-hour quota windows — we can run, hit a limit, come back later, and continue exactly where we left off.

LLM-as-Judge Tuning

Early judge runs showed a consistent problem: the judge was **too strict on form**. It penalised answers for wording differences, for alternative tool paths, and for not citing systems in an exact way — even when the factual content was correct.

We reworked the judge prompt around a new principle: **flexible on form, strict on facts**.

The judge penalises	The judge now accepts
Wrong numbers, IDs, statuses, dates	Alternative wording (“450 THB” = “B450”)
Fabricated data (anything not in seed)	Different tool call paths leading to the same answer
Missing critical facts from the scenario	Summary-level answers when all criticals are present
Internally contradictory reasoning	Extra analysis or context beyond what was asked

- We also redesigned the scoring dimensions and added new structured judge outputs:
- **4 scoring dimensions** (instead of 5): factual accuracy (40%), completeness on critical facts (30%), no hallucination (20%), reasoning clarity (10%).
 - **Critical facts tier** — each scenario defines which facts are critical vs. nice-to-have. An answer passes only if all critical facts are present.
 - **Failure categorisation** — when an answer fails, the judge classifies it: `missed_fact`, `wrong_value`, `hallucination`, `bad_reasoning`, or `refused`. This lets us analyse failure modes in aggregate.
 - **Functional consistency** — in addition to numeric score stability, the judge checks if the high-level conclusion matches across repeated runs of the same scenario.

Evaluation Runs Launched

We ran the full pipeline several times across the week to:

- Validate that resumability works as intended after quota interruptions.
- Confirm the judge scores are now aligned with our manual qualitative evaluation.
- Collect enough runs to make aggregation meaningful.

Each full pass is **27 scenarios × 3 runs = 81 agent evaluations + 81 judge evaluations**.

With the new scenarios covering L1–L5 and five use cases, the result dataset is rich enough to support the analysis planned for the final report.

In parallel we added supplementary metrics captured during each run so the aggregation step can produce meaningful charts:

- Tools actually used and systems actually touched (not just a count).
- Time saved versus the human baseline estimate per scenario.
- Tool efficiency (realistic minimum vs. actual tool calls).
- System coverage ratio (expected systems hit vs. total).

Final Report Drafting

We started drafting the final report in parallel with the evaluation runs. The current draft covers the problem statement, the literature review, the methodology, and the architecture — all sections that are stable regardless of the final numbers. The results, analysis, and conclusion sections are placeholders waiting for the aggregated evaluation data.

The aggregation script is ready and has been tested on partial data. Once the full run completes, generating the final report’s quantitative sections will be a matter of running the aggregator and plugging the numbers into the existing narrative.